

Paper Title

Name, Surname
Student ID

Remarks:

- Do not change the order of slides and remove any slides.
- In case you could not answer a certain question, just mention that you did not answer and leave it blank.
- In case you need more than 1 slide per question, please duplicate the slide and extend the slide number in the title as A/B/C. (example Q1.2A, Q1.2B etc)

Q1.Introduction

- Identify the issues addressed by the assigned paper and identify the main techniques used to address these issues. (5 points)
- Support your analysis with figures/block diagrams.

Q2 Matlab model and List of coefficients

- Include a block level diagram of the modulator and list coefficients.
- Describe how you have derived the coefficients of the modulator.

Q2.1 Proposed technique

- Enable and disable the proposed techniques in the paper. Quantify and compare the results.

Q2.2 Voltage scaling

- Scale the voltage levels to match the paper. What are your assumptions?

Q2.3 Thermal noise and output spectrum

- Add thermal noise and scale the output spectrum like the paper.

Q2.4 Decimation Filter

- Explain the decimation filter and plot the output spectrum before and after decimation filter. Also include transient waveforms.

Q2.5 List of simulation results

- Create an overview of simulation and measurement results presented in the paper in table format and high light what you have analyzed. If not, please motive your reasoning.
- Summarize the assumptions for your matlab simulation setup.

Q2.6-X Matlab simulation comparison

- Describe what is simulated. Use as many slides as necessary to support your analysis. 1 page per simulation. Do not change the layout. Use the proposed formatting below:

Published result

Your result

- Quantify your comparison.

Q3.1 Circuit Non-idealities: Finite amplifier gain & BW

Simulation result. Use the same aspect ratio of this box and be consistent with your formatting of axis and linewidth. The axis should be clearly readable.

- What is the amplifier gain and bandwidth reported in the paper? What did you define in your simulations?
- Explain how introduced non-ideality impacts the performance.

Q3.2 Circuit Non-idealities: DAC mismatch

Simulation result.

Use the same aspect ratio of this box and be consistent with your formatting of axis and linewidth. The axis should be clearly readable.

- Explain how introduced non-ideality impacts the performance. How did you chose DAC mismatch?

Q3.3 Circuit Non-idealities: Excess Loop Delay (ELD)

Simulation result. Use the same aspect ratio of this box and be consistent with your formatting of axis and linewidth. The axis should be clearly readable.

- Describe how did you model the excess loop delay?
- Explain how introduced non-ideality impacts the performance.

Q3.4 Circuit Non-idealities: Clock jitter

Simulation result. Use the same aspect ratio of this box and be consistent with your formatting of axis and linewidth. The axis should be clearly readable.

- Describe how did you model clock jitter ?
- Explain how introduced non-ideality impacts the performance.
- Did the design employ specific techniques to reduce the jitter sensitivity?

Q3.5 Circuit Non-idealities: non-linearity of integrator

Simulation result. Use the same aspect ratio of this box and be consistent with your formatting of axis and linewidth. The axis should be clearly readable.

- What is the non-linearity of the integrator reported in the paper? Which integrator is the dominant one?
- How did you model non-linearity of the integrator
- Explain how introduced non-ideality impacts the performance.

Q4.1-X Circuit Implementation – Block 1

Include a schematic of the block.

Some candidates: DAC, amplifier, comparator, sample/hold

Only describe 1 block per slide and

- Describe the operation of the circuit, and in case of a SC circuit draw different phases and explain the mode of operation.

Q5. Identify and list their weakness and strengths of the main techniques used in the SD modulator architecture. Assess if the adopted technique solves the initial issue. (5 points)

Q5. Strength and weakness of the design

- Identify and list the strength and weakness of the design in a table format.
- Motivate your conclusions if proposed technique solves the initial issue.
- Incase you multiple slides to support your conclusions.

Q6. Compare the adopted techniques to other SD modulator architectures and to different solutions. The comparison should include (but is not limited to) the other works cited in the paper. Prepare or extend the comparison table present in the paper with your analysis (5 points)

Q6. Comparison

- Include an extended version of the comparison table.
- Compare the adopted techniques to other SD modulator architectures and to different solutions.

Q7. Propose improvements to SD modulator proposed in the paper (5 points) and prove your proposed improvement by Matlab simulations. (5 points)

Q7. Proposed Improvements

- Propose improvements to SD modulator proposed in the paper.
- Propose an updated architectural drawing, or circuit diagram.
- Prove your proposed improvement by Matlab simulations.

Summary

- Conclude on your assessment of the paper.